Nano-payments for supervised machine spending

PennyPilot

A spend-control card for AI finance agents buying paid financial facts through Circle Gateway x402, with receipts anchored on Arc.

- Gateway x402
- Arc receipts
- 50 tx proof

$0.003885

average paid action in the compact review

$0.25	$0.006	50
task budget	max per call	receipt tx

pennypilot-five.vercel.app

Contract 0x4c8E...3047

Tiny spend needs control

PennyPilot turns nano-payments into supervised procurement for AI agents.

AI agents can buy useful facts, but the payment rail needs guardrails.

PennyPilot is built for finance workflows where many tiny paid API calls are valuable only if every call is bounded, signed, priced, and auditable.

Too small for cards

Each proof can cost fractions of a cent. Traditional payment overhead destroys the unit economics.

Too many to approve

A useful agent may need several checks in one task. One approval per proof becomes a UX tax.

Too risky without receipts

Finance teams need to know what was bought, from whom, why, and under which policy.

The solution: batch-controlled agent spend

The app keeps the agent useful while avoiding approval fatigue.

One user authorization can cover a bounded batch of paid proofs.

The user sets the task budget, per-call cap, provider allowlist, and proof count. The agent can spend only inside that policy.

Task budget	$0.25 USDC
Per-call cap	$0.006 max per proof
Batch control	user selects how many proofs the agent may buy
Receipt anchor	provider-signed batch recorded on Arc

1 signature -> **6 paid proofs**

Compact review example

ProofDesk	$0.0030	KybTrail	$0.0045
ClearList	$0.0038	AccountLock	$0.0046
FraudLens	$0.0051	QuoteMesh	$0.0022

$0.023308
total batch spend
under budget

$0.003885
avg paid action
below $0.01

Live app demo flow

pennypilot-five.vercel.app

Payment first, proof next

Circle Gateway handles payment authorization; Arc keeps the audit trail verifiable.

Policy

Budget, cap, expiry, provider allowlist, and proof count are set before spend.

->

x402

Paid API endpoint returns PAYMENT-REQUIRED for the selected batch.

->

Gateway

Wallet signs once. Circle Gateway verifies the payment authorization.

->

Provider

Each proof returns data plus a provider-signed receipt.

->

Arc

PennyPilot anchors the receipt batch on the user's spend card.

What judges should watch in the video

Circle-side payment authorization, then Arc-side receipt verification. The key product change is one signature for the selected proof batch.

recordSpendsWithProviderSignatures

Proof package for the judges

A short video can show the live app, Circle flow, Arc explorer proof, and verifier commands.

Live, deployed, and verifier-backed.

The submission is not just a mock UI: the repo includes proof scripts that read Arc Testnet and confirm bytecode, policy caps, receipt replay protection, provider spend, and digest equality.

Live app	pennypilot-five.vercel.app
GitHub	github.com/Kxazar/pennypilot
Contract	0x4c8E...3047
50 tx proof	0x9f65...8ee7

50

verified receipt transactions

Arc Testnet proof

<= $0.01

paid action requirement

provider prices stay below the cap

1

batch user authorization

selected proof count, one signature

test | arc proof | 50 tx proof